



# OPEN Avian neighbours: density patterns of synanthropic birds along a rural–urban landscape gradient in Northern India

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Unbiased and accurate estimation of bird density are prerequisites to monitor the impact of urbanization on avian communities. Synanthropic birds are reliable indicators of landscape modification in small tropical cities with rural–urban ecological settings. We conducted 183 fixed-width point counts to record avian communities along the rural–urban gradient in Mirzapur and Bhadohi. We applied hierarchical distance-sampling to estimate the summer density of 35 bird species across eight foraging guilds in response to vegetation, land cover, human activity and housing type, accounting for detection probability as a function of weather. Twenty-seven species (77%) showed differences in density across the landscape gradient. The density of frugivores was highest in urban gradient, a carnivore was highest in rural, and a scavenger was highest in the semi-urban gradient thereby supporting the resource concentration hypothesis. Insectivores, granivores and omnivores showed variable density patterns along the gradient. The relatively lower density of synanthropic birds in semi-urban and urban fringes indicates the need for enhanced green space and vegetation structure along intermediate landscape gradient. The habitat associations and population sizes of synanthropic birds are useful for landscape managers and local stakeholders to maximize avifaunal diversity in the intermediate landscapes of the rural–urban continuum in Uttar Pradesh.

**Keywords** Gradient ecology, Hierarchical distance sampling, Intermediate landscape, Point count surveys, Uttar Pradesh, Avian community, Rural–urban continuum

Alterations in biotic-abiotic environments and loss of avian habitat due to the fragmentation and increasing urbanization provide novel challenges to the persistence of synanthropic birds. Urbanization transforms the natural vegetation and community structure<sup>1</sup>, thereby impacting biodiversity through habitat loss, fragmentation, green space decline, and aids biotic homogenization in the ecosystems<sup>2,3</sup>. The vertical and horizontal heterogeneity in urban, semi-urban and rural ecological settings can accommodate sympatric synanthropic bird species with niche partitioning<sup>4,5</sup>. Studies have shown variable responses of synanthropic birds to the increasing urbanization at regional scales<sup>6–8</sup>. Landscape modification associated with urban growth has been impactful in the last five decades causing not only extensive biodiversity loss but also environmental degradation, poor air quality with serious consequences on human health<sup>9–12</sup>. It involves exchange of native vegetation and croplands with impervious ground layer and artificial infrastructure and increased human population density<sup>13</sup>. The emerging urban landscapes depicts complex, heterogeneous patches of distinct land use, comprising educational institutions, industrial zones, shopping complex, residential buildings, park, gardens, green roofs, road, railway, ponds and water canals. Urban areas with high bird diversity can provide several ecosystem services<sup>14</sup>.

In city landscapes, the adequate proportion of green and blue space can provide sustainable resources for both resident and migratory synanthropic birds<sup>15</sup>. The association of habitat heterogeneity with synanthropic bird communities can be understood better when bird populations are monitored in the non-migratory season i.e. summer to study the impacts of urbanization on the densities of “urban adaptor” and “urban avoider” bird species<sup>7,16,17</sup>. Accurate density estimates enable ecologists and conservationists to focus on species sensitive to urban planning and development<sup>18</sup>. Availability of green spaces and multi-vegetative stratum facilitates

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resource niche partitioning among several sympatric bird species of various foraging guilds<sup>19,20</sup>. Population-level patterns of several partial-synanthropic and synanthropic bird species along the rural–urban gradient are poorly understood<sup>8,21</sup>. To test the resource concentration hypothesis in synanthropic birds along the rural–urban gradient<sup>22</sup>, we expect that the density of bird species should increase in the landscape gradient as a result of greater concentration of resources. Hence, assessing the relationships between bird abundance with the local-scale vegetation and landscape properties such as housing type, human activities, and land cover provides a better understanding of species-specific requirements of low- and high-quality habitat patches along the rural–urban continuum. Bird species respond differentially to the gradients of housing density and green spaces. For instance, a lower density of “urban adapter” or “urban exploiter” could indicate signals of ecological stress especially when territories are established in low-quality habitat patches along the rural–urban gradient<sup>23</sup>. Hence, birds are effective sentinels in rural–urban ecological settings due to their close association with human activities. Their site-specific density along rural–urban gradients can be treated as early-warning signals of sensitivity to the changes in habitat quality and environmental conditions<sup>24</sup>.

In India, the rise in economic development and expansion of urban land use has grown rapidly spreading across many natural green spaces in rural and peri-urban areas<sup>25</sup>. Particularly, summer temperatures in Northern India have increased<sup>26</sup> and the environmental conditions in Indian summers are challenging as birds rely on the remnant green cover available in the urban–rural land use matrix. Nearly one-third of India's human population resides in urban areas and future projections indicate exponential growth<sup>27,28</sup> thereby exerting unprecedented pressure on natural ecosystems, leading to habitat fragmentation<sup>29,30</sup>, biodiversity loss<sup>31</sup>, and environmental degradation<sup>32,33</sup>. Moreover, the pace of urbanization has outstripped the development of sustainable urban planning strategies, creating significant gaps in integrating ecological principles into city-growth models. Understanding this dynamic is crucial for mitigating ecological impacts and promoting biodiversity conservation in rapidly urbanizing cities in India. Studies on bird diversity, richness and density has been conducted in wetlands<sup>34</sup>, farmlands<sup>35</sup>, forests<sup>36,37</sup> and grasslands<sup>38</sup>, yet explicit investigations focussing on the density patterns along the rural–urban continuum remains limited in India<sup>21,39,40</sup>. Some studies employed point counts and estimated density using Reynold's formula whereas others applied line transect survey method and estimated density as number of individuals/ (transect length in km  $\times$  effective transect width in km). In contrast, some have applied distance sampling approach for bird density estimation<sup>41–43</sup>. Hierarchical distance sampling allows researchers to explicitly model both the biological process (true density) and the observation process (variability in detection probability) simultaneously allowing more accurate and robust density estimates.

The sparsity of robust density estimates of several common rural–urban birds brings us to explore how the synanthropic bird species density varies in rapidly urbanizing cities of Mirzapur and Bhadohi districts in Uttar Pradesh, Northern India. Using 183 fixed-width point count sampling method, we applied hierarchical distance sampling (HDS) models to test the influence of landscape properties, human activity, housing type and local vegetation on density estimation of 35 bird species, while accounting for variation in detection probability as a function of weather parameters. Our objectives were to: (i) assess the influence of local-habitat and landscape factors on the density of synanthropic bird species belonging to different feeding guilds; and (ii) compare bird species density along the urban, semi-urban and rural landscape gradient. In human-dominated environments we expect that food resources are generally abundant for bird species adapted to rural–urban gradients. The local vegetation, anthropogenic activity and land cover are important determinants of the abundance and quality of food resources and may vary spatially along the rural–urban gradient. These resources can be used by birds adapted to either of the three gradients where they may reach high densities. We predict a gradual or monotonous increase or decrease in bird density from rural to urban areas or vice versa in select foraging guilds. We predict that select bird species reaching higher densities in rural, semi-urban or urban are assumed to be adapted to either of these gradients. We predict that density patterns would differ among sympatric avifaunal groups through niche partitioning.

## Results

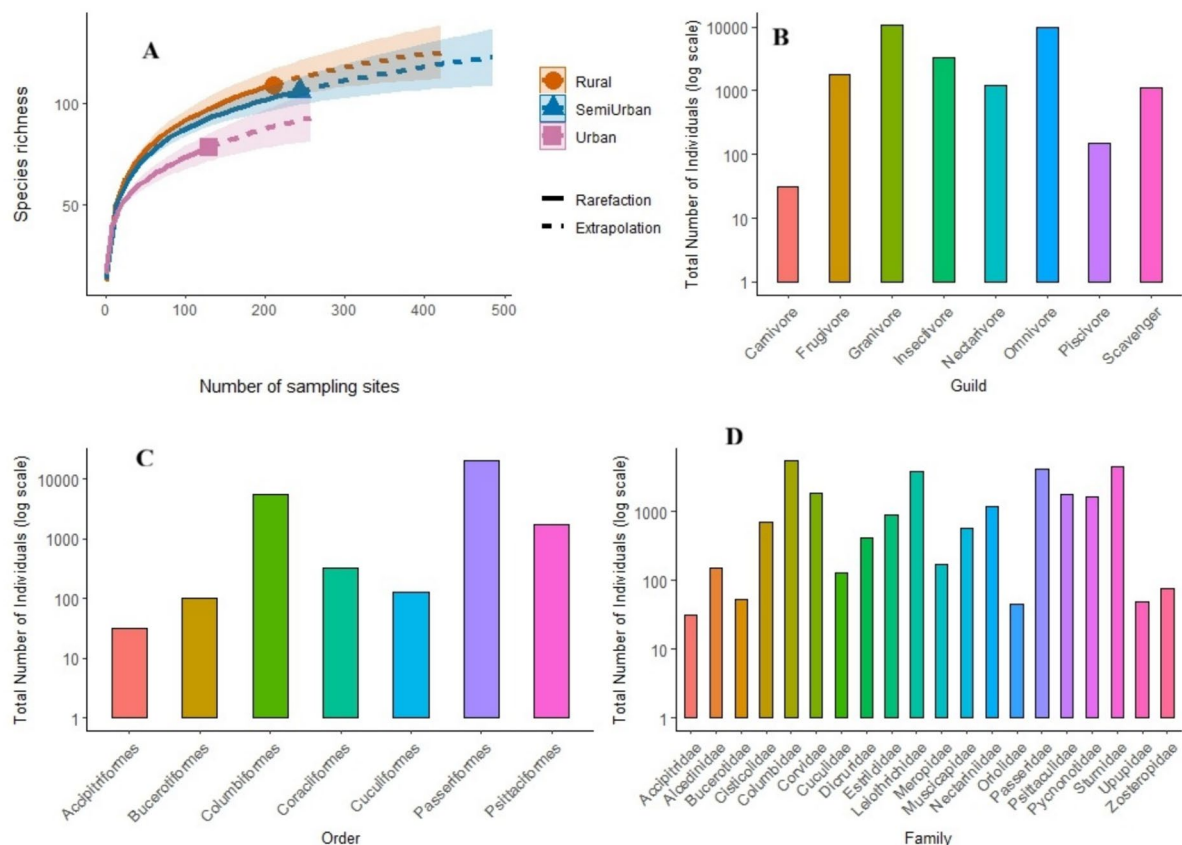
### Avian community composition and habitat structure

A total of 27,818 individuals belonging to 35 bird species were recorded during the survey. Species comprised of 8 foraging guilds, 7 orders and 20 families (Fig. 1). The rarefaction curves of the three landscape gradients showed that species richness increased with the number of sites sampled, eventually reaching an asymptote. The urban gradient reached an asymptote earlier, compared to the rural and semi-urban gradients. Chao estimation indicated that sampling captured approximately 88% of the potential species richness in urban areas (mean Chao =  $88.56 \pm 7.67$  SE), 80% in semi-urban areas (mean Chao =  $131.78 \pm 16.15$  SE), and 84% in rural areas (mean Chao =  $129.64 \pm 13.48$  SE), suggesting the achievement of sufficient sampling effort.

The NMDS plot (stress = 0.06) indicated good fit for the ordination and showed 25% of bird species overlapping rural–urban gradient; semi-urban transitioned (Fig. 2). The stress value was within acceptable limits, suggesting a reliable two-dimensional representation of the state variables and landscape gradient. The ordination revealed clustering of sampled point count stations along the rural–urban gradient, showing the rural point count stations on one end and urban stations on the other while the semi-urban stations occupied the intermediate position. Among the variables, proximity to water bodies ( $R^2 = 0.048$ ,  $P < 0.05$ ), tree count ( $R^2 = 0.054$ ,  $P < 0.05$ ), and the prevalence of thatched-roof housing type ( $R^2 = 0.11$ ,  $P < 0.05$ ) showed significant association with rural areas, reflecting more natural and less modified landscapes. In contrast, the presence of multi-storey buildings ( $R^2 = 0.20$ ,  $P < 0.05$ ) showed a strong association with urban areas, highlighting increased built-up infrastructure.

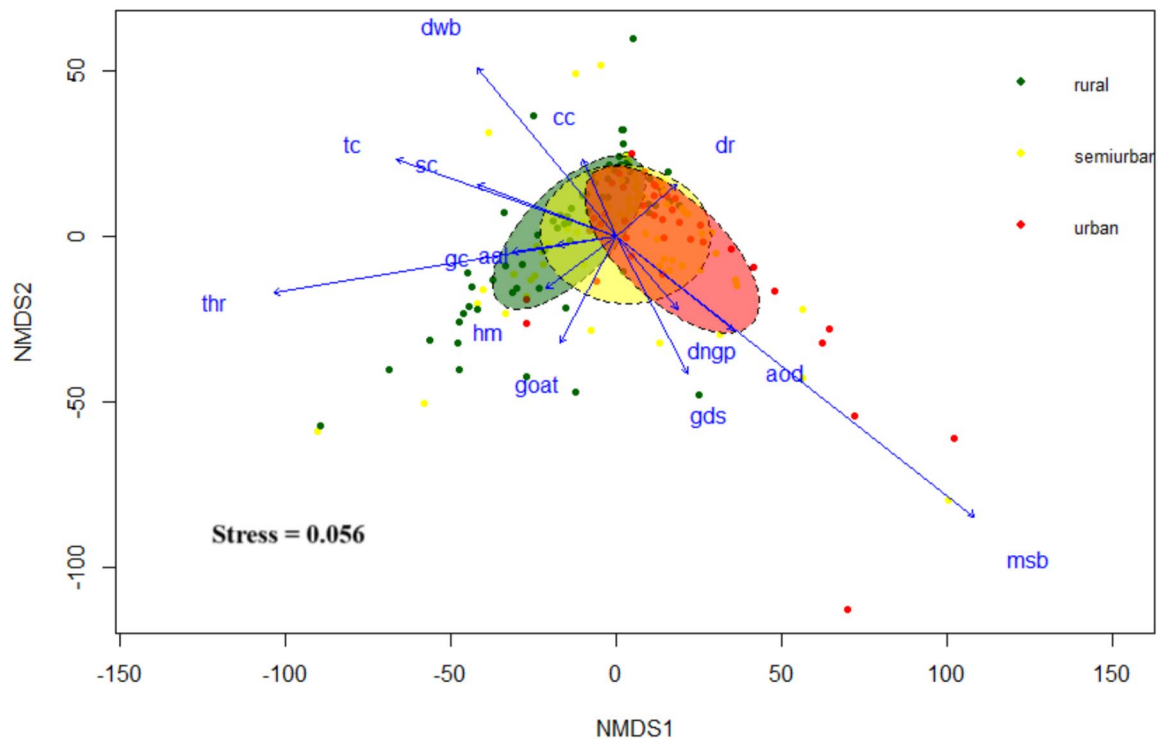
### Detection probabilities of bird species

Among the 35 bird species, House Sparrow (*Passer domesticus*, 720 detections), Rock Pigeon (*Columba livia*, 643), and Common Myna (*Acridotheres tristis*, 567) were the most frequently detected species. Red-vented



**Fig. 1.** Rarefaction curves depicting species accumulation by sampling effort (A), Total number of individuals across avian guilds (B), order (C) and family (D) according to AVONET traits.

Bulbul (*Pycnonotus cafer*), Indian Pied Starling (*Gracupica contra*), Jungle Babbler (*Turdoides striata*), and Rose-ringed Parakeet (*Psittacula krameri*) also showed high detections (>300). Medium detections (100–300) were recorded for Large Grey Babbler (*Turdoides malcolmi*), Black Drongo (*Dicrurus macrocercus*), and Spotted Dove (*Spilopelia chinensis*). In contrast, species with low detections (<35), including Indian Grey Hornbill (*Ocyrocus birostris*), Eurasian Hoopoe (*Upupa epops*), Shikra (*Accipiter badius*), Indian Golden Oriole (*Oriolus kundoo*), and Scaly-breasted Munia (*Lonchura punctulata*). The half-normal detection function was the best fit for 49% (N = 17), hazard-rate for 37% (N = 13), and exponential for 14% (N = 5). The GOF tests for Indian Grey Hornbill and Eurasian Hoopoe global models showed that the state variables in the top models were not significant (Table S1). AQI emerged as the most common detection covariate influencing detectability of 19 species (54%), indicating a strong association between air quality and detection probability of many urban and semi-urban bird species. Wind speed influenced the detection probability of 14 species (40%), while temperature was influential for 13 species (37%). Twelve species showed detectability influenced by a combination of temperature and AQI. Oriental Magpie Robin (*Copsychus saularis*) and Plum-headed Parakeet (*Psittacula cyanocephala*) were not influenced by any weather parameter. Rose-ringed Parakeet, Eurasian Hoopoe, Large Grey Babbler, Red-vented Bulbul, Indian Golden Oriole, Greater Coucal (*Centropus sinensis*), Rufous Treepie (*Dendrocitta vagabunda*), Indian Silverbill (*Euodice malabarica*), Spotted Dove and Shikra detections increased with AQI. While detection probability of Ashy Prinia (*Prinia socialis*), Common Tailorbird (*Orthotomus sutorius*), Brown Rock Chat (*Oenanthe fusca*), Large-billed Crow (*Corvus macrorhynchos*), Jungle Babbler, Indian Pied Starling, Brahminy Starling (*Sturnia pagodarum*), and Eurasian Collared-Dove (*Streptopelia decaocto*) declined with increasing AQI. The detections of Purple Sunbird (*Cinnyris asiaticus*), Asian Green Bee-eater (*Merops orientalis*), Black Drongo, Common Myna, Indian White-eye (*Zosterops palpebrosus*), Indian Silverbill, House Sparrow and White-throated Kingfisher (*Halcyon smyrnensis*) declined with wind speed while Common Tailorbird and Eurasian Collared-Dove showed humped-shaped relationship with wind speed. The detections of Brown Rock Chat, Bank Myna (*Acridotheres ginginianus*), and Rock Pigeon increased with wind speed. The detections of Indian Grey Hornbill, Ashy Prinia, Indian Robin (*Saxicoloides fulicatus*), Large Grey Babbler, Red-whiskered Bulbul (*Pycnonotus jocosus*), Indian Silverbill, Spotted Dove, Shikra, House Crow (*Corvus splendens*) and White-throated Kingfisher declined with increasing temperature. In contrast, Brahminy Starling and Laughing Dove (*Spilopelia senegalensis*) detections increased with temperature (Fig. S2A–E).



**Fig. 2.** A non-metric multidimensional scaling (NMDS) ordination of 183-point count stations based on vegetation, landscape features, human activity and housing density. NMDS using Euclidean method for urban (red ellipse), semi-urban (yellow ellipse), and rural (green ellipse) gradients with respective coloured dots and the state variables axis (blue lines). See Table 2 for the descriptions and abbreviations of variable names.

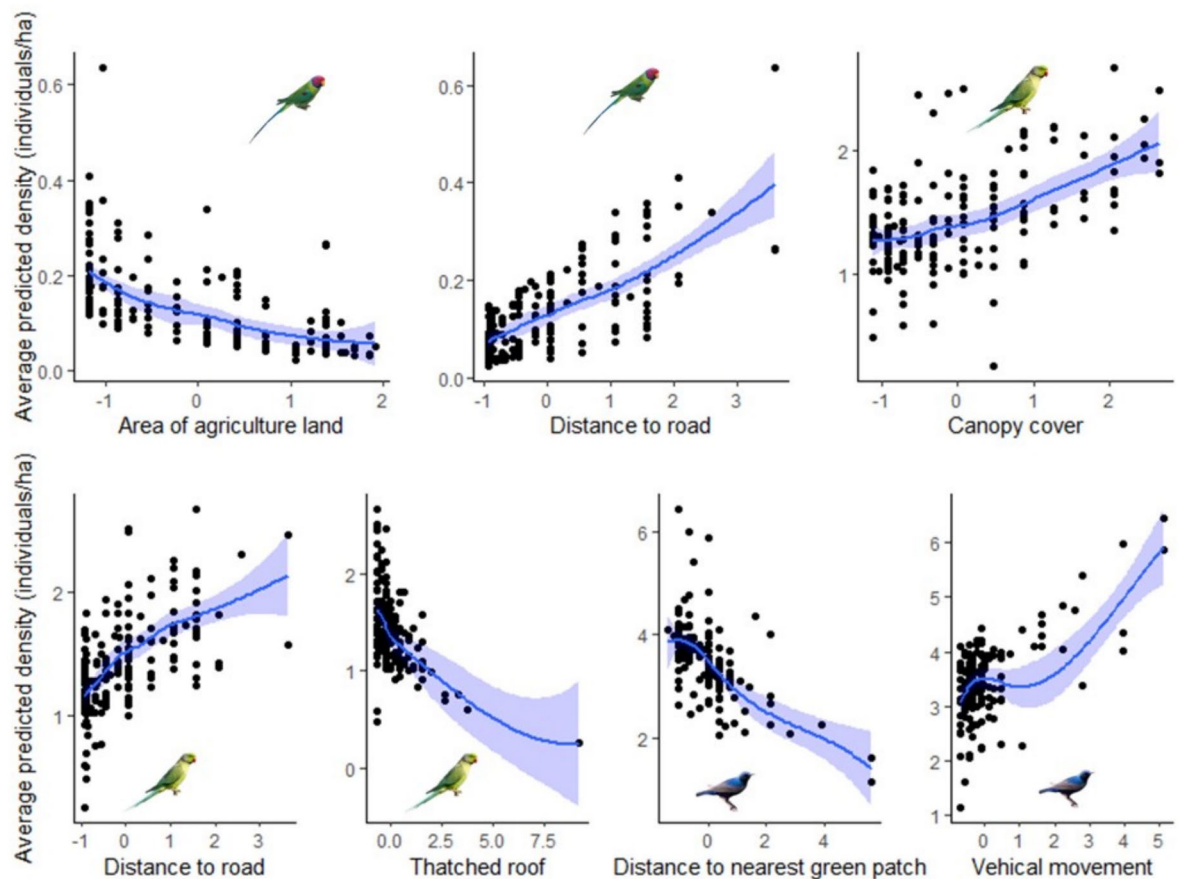
### Determinants of synanthropic bird species density

Variables related to vegetation structure (canopy, shrub, grass cover, tree count), proximity to water bodies, and built-up elements (roof type, building density, garbage dumps) frequently appeared in top models. The  $\Delta AIC_w$  for all species ranged from a minimum of 0.82 to 0.99 (Table S2). Certain covariates such as “distance to road” or “distance to nearest green patch” showed species-specific relevance, underlining the importance of ecological niche and tolerance to disturbance (Table S3). The predicted density of Ashy Prinia increased with agriculture area, while density of Plum-headed Parakeet, Oriental Magpie Robin, Indian Robin, Red-whiskered Bulbul, Jungle Babbler and Bank Myna decreased with agriculture area. The predicted density of Plum-headed Parakeet, Rose-ringed Parakeet, Spotted Dove increased with distance to the nearest road while Asian Green Bee-eater, Brahminy Starling and House Crow decreased with proximity to road. Rose-ringed Parakeet, Common Tailorbird, Oriental Magpie Robin, Indian Golden Oriole, Indian Silverbill, Purple Sunbird, Rock Pigeon, and White-throated Kingfisher density was associated with canopy cover, number of trees and green patch. The predicted density of Brown Rock Chat, Asian Green Bee-eater, Black Drongo, Indian White-eye, Large-billed Crow, Scaly-breasted Munia decreased with number of trees, and canopy cover. Laughing dove had a weak association with grass and shrub cover while Ashy Prinia had a stronger positive relationship with shrub cover. Predicted density of Common Myna increased with grass cover. The predicted density of both species of parakeets and House Crow increased in proximate distance to road, while Asian Green Bee-eater weakly declined in proximate distances while density of Spotted Dove and Brahminy Starling was higher away from roads. Density of Rose-ringed Parakeet, Ashy Prinia, Common Tailorbird, Oriental-magpie Robin, Jungle Babbler, Bank Myna, Eurasian Collared-Dove, both species of bulbuls, decreased with increasing number of thatched-roof houses. House Sparrow and House Crow densities increased with number of thatched-roof houses. Predicted density of Purple Sunbird increased with vehicular movement while Indian Golden Oriole, Bank Myna and Brahminy Starling declined with increasing vehicular movement. Asian Green Bee-eater and Oriental Magpie Robin declined with increasing distance to water body (Figs. 3, 4, 5, 6, 7, Table S3).

### Bird species density and the total population size along the rural–urban gradient

Among frugivores and nectarivores, the Purple Sunbird had the highest population ( $N_{hat} = 2002.18$ ), while the Indian Grey Hornbill had the lowest ( $N_{hat} = 38.15$ ). Within the insectivore guild, the Indian Robin had the highest population ( $N_{hat} = 995.97$ ), contrasting with the Eurasian Hoopoe, which had the lowest population ( $N_{hat} = 128.59$ ). The omnivore guild was led by the Asian Pied Starling ( $N_{hat} = 8222.55$ ) and the Indian Golden Oriole had the lowest population ( $N_{hat} = 67.95$ ). Among granivores, House Sparrow had the highest population ( $N_{hat} = 7681.65$ ), with the Indian Silverbill being the least numerous ( $N_{hat} = 192.15$ ). Lastly, in the carnivore,





**Fig. 3.** Predictions of density for parakeets and Purple Sunbird generated from model averaged coefficients of significant state variables in the top HDS models. The state variables held at their mean values (solid blue line) and the confidence interval (shaded region) are shown.

piscivore and scavenger guild, the House Crow had the highest population ( $N_{\text{hat}} = 655.93$ ), while Shikra was the lowest ( $N_{\text{hat}} = 124.06$ , Table S4).

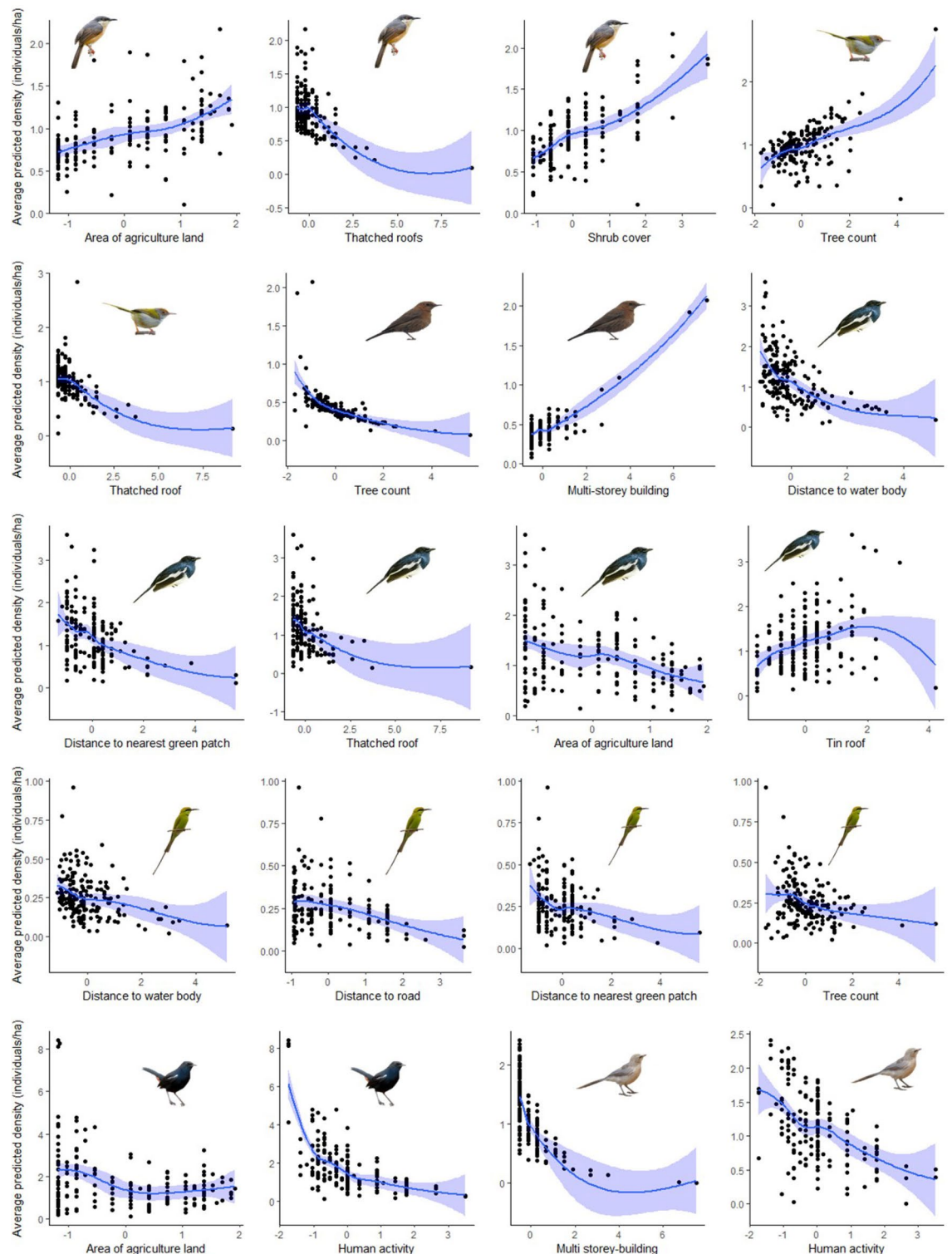
Overall, the density of eight species did not vary significantly along the rural–urban gradient while the remaining 27 species showed significant variation. Within the guild, the predicted density of frugivorous bird species was found to significantly differ ( $P < 0.05$ ) across the gradient, being highest in the urban and lowest in rural. The predicted density of Purple Sunbird, the only nectarivorous bird was the highest in semi-urban landscape. The predicted density of eight insectivorous bird species was found to significantly differ ( $P < 0.05$ ) along the gradient. We found that 38% of the insectivorous bird species density was higher in urban, 25% were higher in rural and 13% were higher in semi-urban landscape. The predicted density of 46.15% of the omnivorous bird species were found to be higher in urban and 31% were higher in rural landscape, whereas three omnivorous birds did not vary significantly ( $P > 0.05$ ) along the gradient. Although 57% of the granivores differed significantly across the gradient, nearly 43% did not vary significantly. The density of 29% of the bird species was higher in rural areas and 29% were higher in urban areas. The density of Shikra and House Crow varied significantly across the landscape gradient ( $P < 0.05$ ), whereas White-throated Kingfisher did not vary significantly (Fig. S3–S7, Table S5).

### Spatial patterns of site-abundance of bird species

Resident bird species showed distinct spatial patterns of site-level abundance, allowing classification into low, medium, and high abundance groups. High-abundance species, comprised 35% of the total, medium-abundance species, comprised 38% and low-abundance species comprised of 27%, restricted to few sites, possibly due to specialized habitat requirements. The site-level abundance of 13 high-abundance species had maximum predicted density ranging between 4.21 to 27.47 ind/ha. The site-level abundance of 11 medium-abundance species had maximum predicted density ranging from 2.05 to 3.59 ind/ha. The site-level abundance of 11 low-abundance species had maximum predicted density ranging from 0.01 to 1.70 ind/ha (Fig. S8A–C).

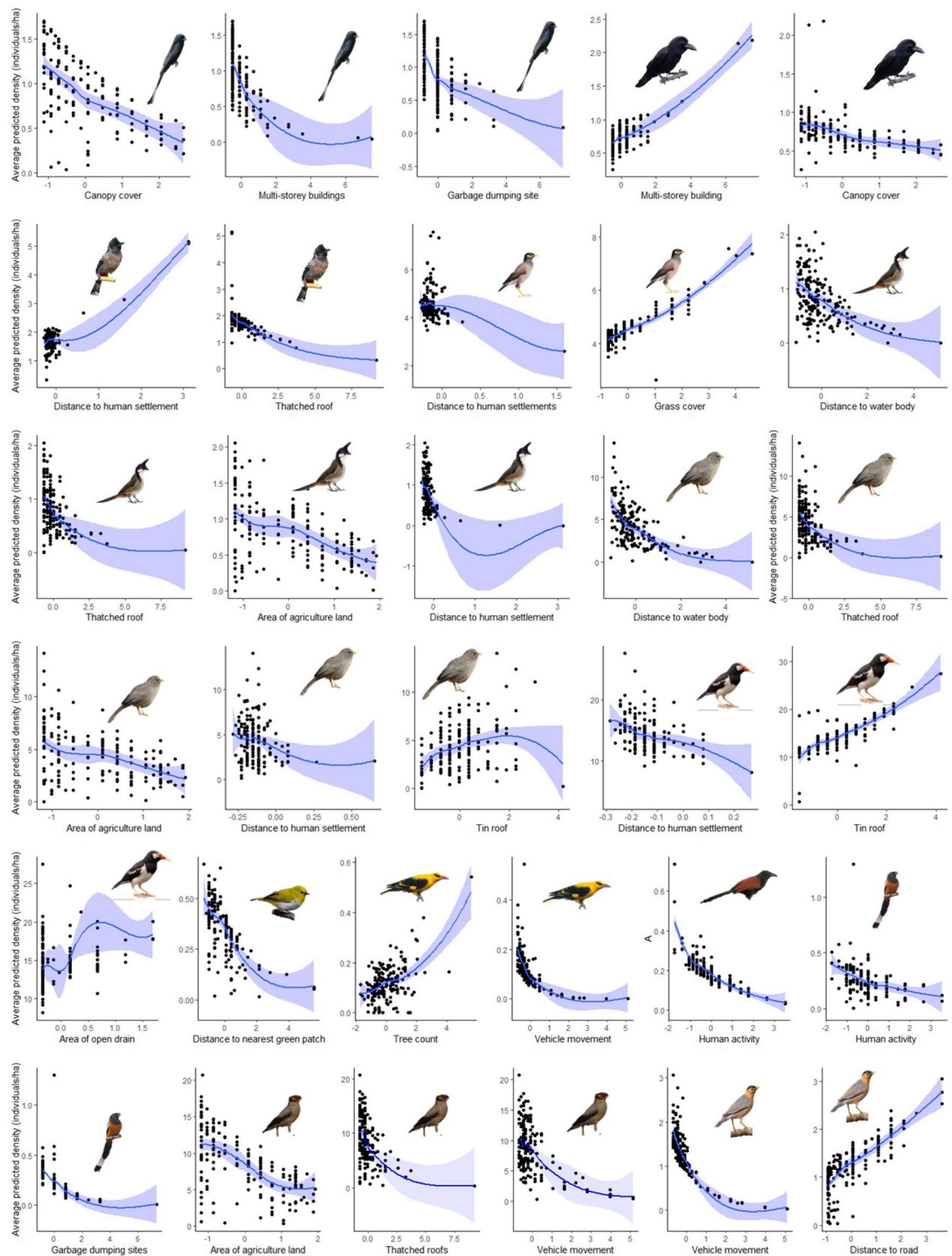
### Discussion

This study is the first to apply a novel hierarchical distance sampling approach to estimate robust density estimates of 35 synanthropic bird species while accounting for detection probability along the rural–urban continuum in



**Fig. 4.** Predictions of density for insectivores generated from model averaged coefficients of significant state variables in the top HDS models. The state variables held at their mean values (solid blue line) and the confidence interval (shaded region) are shown.

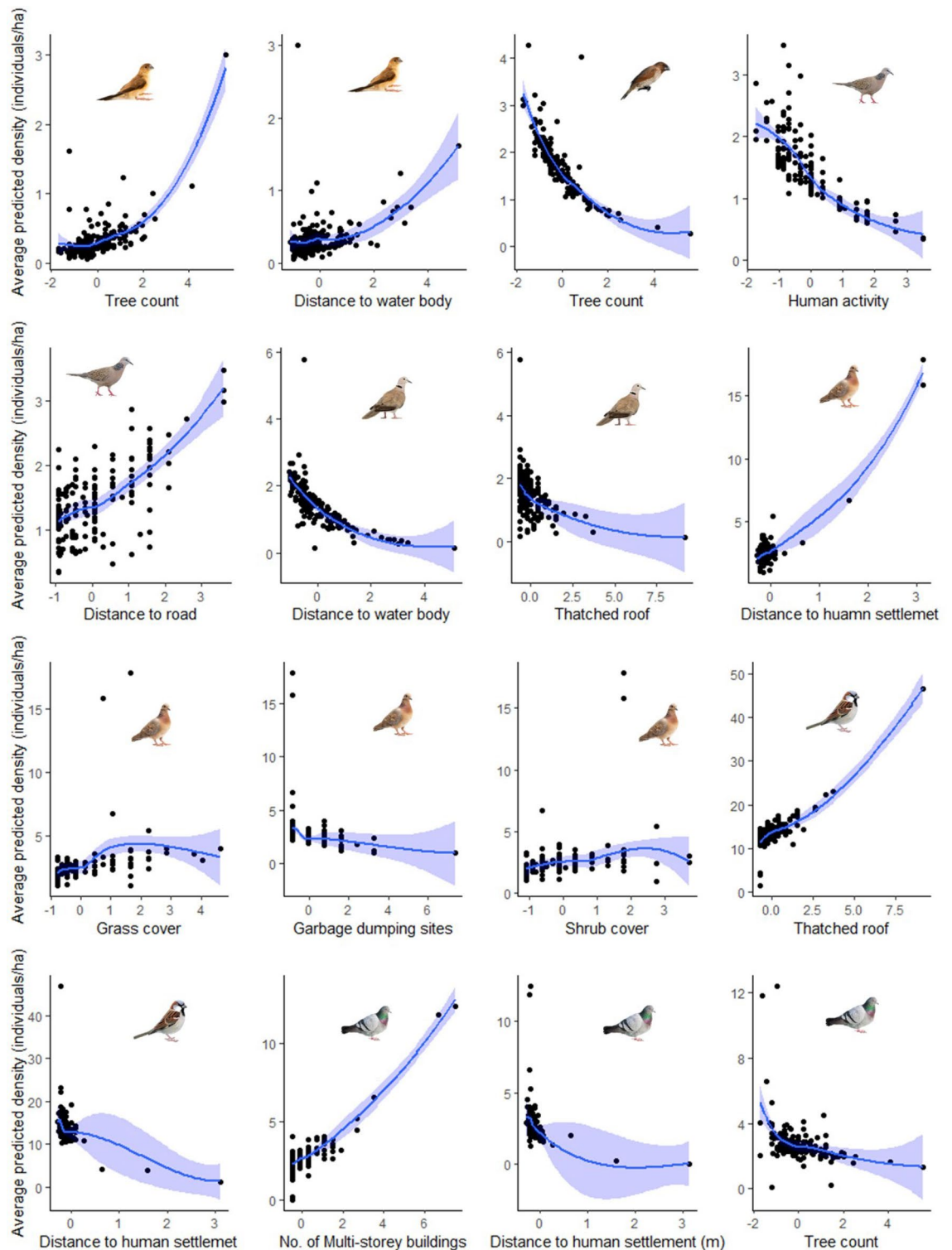
Mirzapur and Bhadohi districts of Northern India. Our models identified the direction and shape of response curves to test the influence of landscape properties i.e. land cover, local vegetation, green patches, housing type and human activities on site-level density of synanthropic birds. We observed five broad patterns of density variation along the rural–urban gradient. The woodland bird density increased from rural to urban gradient, and considerable proportion of insectivores, granivores and omnivores either increased or decreased along the



**Fig. 5.** Predictions of density for omnivores generated from model averaged coefficients of significant state variables in the top HDS models. The state variables held at their mean values (solid blue line) and the confidence interval (shaded region) are shown.

landscape gradient. We found that the NMDS plots showed that sub-urban gradient represents intermediate levels of anthropogenic disturbance<sup>44,45</sup>, where only a few species showed higher or lower densities relative to urban and rural gradient. House Crow and Asian Green Bee-eater showed inverted U-shaped pattern. While Purple Sunbird and Brahminy Starling showed U-shaped pattern at intermediate level of urbanization, there was no significant difference in the density across the gradient. Some species density remained relatively constant

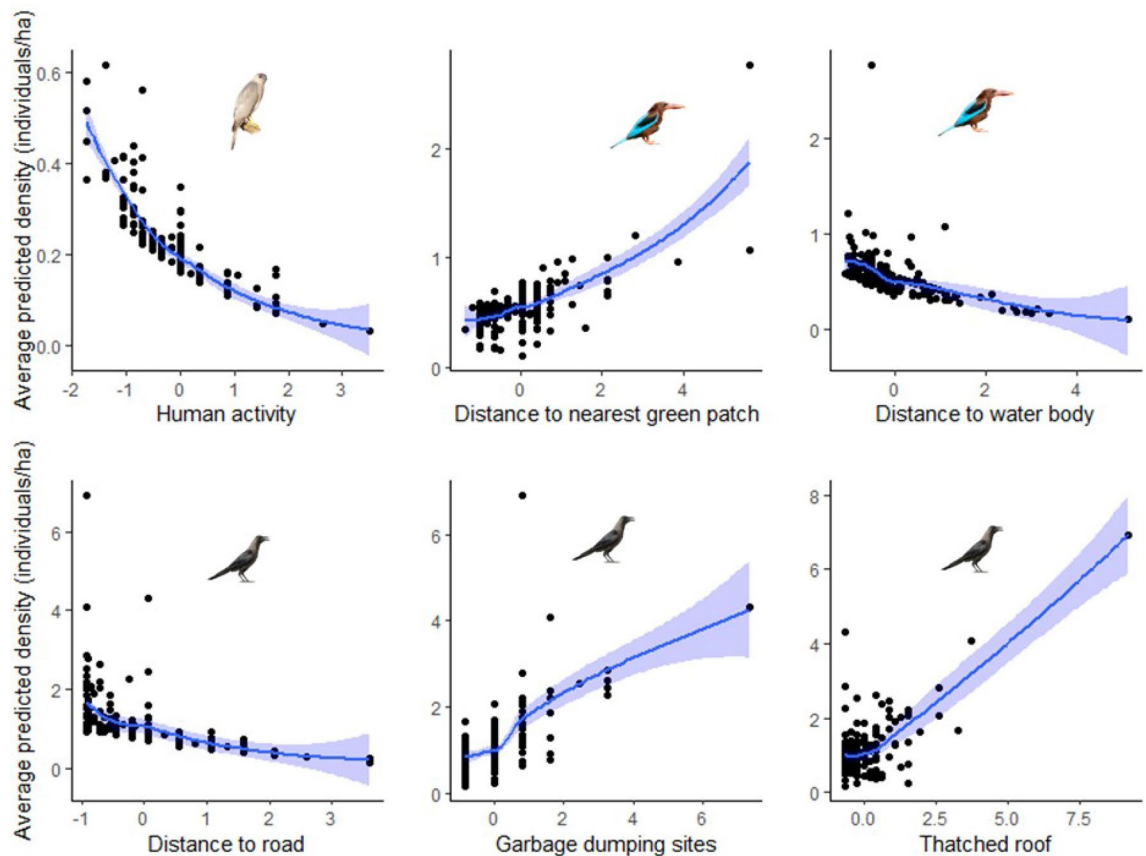




**Fig. 6.** Predictions of density for granivores generated from model averaged coefficients of significant state variables in the top HDS models. The state variables held at their mean values (solid blue line) and the confidence interval (shaded region) are shown.

across the gradient. The variable patterns likely emerge from the interacting effects of resource availability<sup>46</sup> and habitat structural complexity<sup>47</sup> along the rural–urban continuum as reported in earlier studies<sup>8</sup>. The multi-model selection approach revealed that the density of synanthropic birds was best explained by variable combinations of site covariates<sup>48</sup>. Models incorporating covariates performed better than the null model. Cumulative AIC weights varied across species, indicating differing levels of model certainty and variable importance.





**Fig. 7.** Predictions of density for Shikra (carnivore), White-throated Kingfisher (piscivore) and House Crow (scavenger) generated from model averaged coefficients of significant state variables in the top HDS models. The state variables held at their mean values (solid blue line) and the confidence interval (shaded region) are shown.

We found that species detectability varied with weather parameters, where AQI played a dominant role in predicting detection probabilities of many bird species followed by temperature. This shows that more than half of the analysed species were sensitive to environmental variables such as air quality and temperature, which is congruent to other reports<sup>49</sup>. Nonetheless, the moderate AQI values in our study site shows overall good environmental health condition in these districts when compared with other growing metropolitan cities in the state<sup>50,51</sup>. Notably, 52% of the sampled point count stations had moderate AQI and 3% had poor AQI. However, the increased detections of Brahminy Starling and Laughing Dove with temperature could explain species-specific thermal tolerances or adaptive traits<sup>52,53</sup>.

Our site-level estimates suggest that high-abundance and medium-abundance species among sympatric groups partition their niches either through habitat preference, vertical vegetation strata, or spatial avoidance. The low-abundance species showed specialized habitat occurrences, restricted distribution and avoidance of highly anthropized environments, with occasional co-occurrence with sympatric species in transitional habitats. For instance, the Indian Grey Hornbill was primarily associated with well-wooded patches as reported in other urban spaces of Northern India<sup>54,55</sup> and Rose-ringed Parakeet exhibited wide habitat tolerance but was more frequent in areas with tree cover especially near the roadside avenue trees in urban areas as it showed a strong response to both vegetation structure and built-up environments. This corroborates with recent studies from that reported the importance of mature street trees in mitigating negative effects of increasing urbanization in Delhi<sup>56</sup>. Tree cavity and canopy nesting birds such as parakeets and hornbills utilize mature avenue trees in urban spaces<sup>20,54</sup> which could explain their higher density in the urban gradient. Further loss in the remnant mature woodland trees, and cavity-rich habitats in growing urban spaces may limit nesting opportunities for woodland-associated birds<sup>20,57</sup>. We found that most of the insectivorous bird species showed a stronger association with open agricultural areas and shrub-dominated landscapes<sup>58</sup>. An exception was the Brown Rock Chat, which had a closer association with urban built-up environments, maybe because the species exploits nesting opportunities in crevices, and other anthropogenic microhabitats<sup>59</sup>. The mixed density patterns of omnivorous bird species across the rural–urban gradient, reflects their close association with human settlements, likely driven by the availability of anthropogenic food resources, while others appeared sensitive to human activity. Such responses might suggest context-dependent nature of synanthropic bird species responses to urbanization<sup>60–62</sup>. Anthropogenic disturbances such as vehicular traffic and human footprint may impact nesting, and foraging activities of Bank Myna and Brahminy Starling. Notably, the Bank Myna is on a declining trend in the state according to SoIB<sup>63</sup>.

The observed association between housing type and granivorous species may be attributed to the availability of nesting opportunities. Previous studies have similarly reported that traditional thatched roofs provide suitable nesting sites for House Sparrow<sup>64</sup>, while built-up structures are favourable for Rock Pigeons<sup>65</sup>. The higher density of granivores in rural gradient explains the availability of seed resources in rural croplands and fallow lands, where species such as the Indian Silverbill, Spotted Dove, and Laughing Dove thrive<sup>66</sup>, and the proportion of mixed-grass and grain patches sustain the efficient ground-foraging strategies of these granivores<sup>67</sup>. The density estimates of bird species within the carnivore, scavenger, and piscivore guilds could be underrepresented at the guild level due to the few species recorded. Shikra was sensitive to human disturbance, probably due to its predatory behaviour. The understorey shrub cover in urban areas in our study sites might support small-body birds with specialized arboreal nesting habits (e.g. sunbird, tailor birds and white eyes), while the relatively larger ground foragers (e.g. Eurasian Hoopoe, Large Gray Babbler, Greater Coucal) had higher density in rural gradient. The Greater Coucal and Rufous Treepie co-occurred in shrubby or semi-natural areas. A positive relationship was found between ground nesting birds and shrubby habitats in Delhi<sup>40</sup> and other western countries<sup>58</sup>. The loss of native vegetation and lower vegetation complexity in rapidly developing metropolitan cities like Delhi<sup>40</sup> and Bangalore<sup>68</sup> can increase nest predation probability of cats, dogs and corvids<sup>69</sup> and apparent lower survival rates of ground-nesting birds in cities.

The sympatric avian groups; babblers, bulbuls, doves and pigeon, starlings, prinia, tailor bird, bee-eater and drongos showed niche partitioning across space and vegetation stratum. Ashy Prinia and Common Tailorbird were associated with shrub-dominated and tree-dominated habitats, respectively, indicating niche partitioning based on vegetation structure. Ashy Prinia is associated with dense understorey shrubs having high arthropod turnover<sup>70</sup>, while Common Tailorbird depends on taller foliage layers that facilitate leaf-stitch nesting and vertical foraging movements<sup>71</sup>. The Jungle Babbler, Rock Pigeon, and Eurasian Collared-Dove had higher density in urban gradient while the Large Grey Babbler, Spotted Dove, and Laughing Dove had higher density in rural gradient. Red-vented Bulbul seemed to be a generalist while the Red-whiskered Bulbul thrived in gardens and roadside avenue trees, in semi-natural spaces in urban-edge landscapes. Asian Green Bee-eater and Black Drongo partially overlapped in open habitats as aerial foragers<sup>72</sup>. Therefore, nesting and foraging are important intrinsic traits driving bird density variation along the landscape gradient<sup>73–75</sup>.

There are several urban bird community and population studies from large metropolitan cities of Northern India<sup>8,40,76,77</sup>, however avian studies are potentially lacking from rural settings and small growing urban towns. When urban growth takes place at a slow pace and at a lower intensity, consequently, synanthropic birds often show moderate increases in abundance towards urban cores while simultaneously showing stronger associations with mixed habitats and peri-urban or edge habitats<sup>44</sup>. In contrast, Delhi represents a high-intensity urban system characterized by patchy urban forest, extensive impervious surfaces, high human activity, and reduced vegetation heterogeneity. Additionally, while we referenced state-level population trends from the SoIB report using available citizen science data from northwestern Uttar Pradesh, a major data gap persists for the eastern part of the state. Presently, there is no complete coverage of eBird checklist for eastern part of Uttar Pradesh, limiting comprehensive comparisons, therefore our study fulfilled the avifaunal knowledge gap in eastern part of the state in addition to the recent studies conducted in agricultural landscapes<sup>44</sup>. Our findings offer useful insights into the ecological significance and the implications for urban planning in newly emerging cities for avian biodiversity conservation.

## Conclusions and management implications

Reports on land cover change revealed a substantial 25% increase in built-up areas over the past decade in Mirzapur, alongside 8% and 6% decline in barren and fallow lands respectively<sup>78</sup>. This shift indicates rapid urbanization and land-use intensification, with dual impacts on the habitat quality and availability for synanthropic birds. Our findings, along with national trends reported in SoIB<sup>52</sup>, suggest mixed responses to these changes: while 11% of species show increasing population trends, only 3% are declining and 43% remain stable. Our figures highlight that although some species are adapting to human-modified landscapes, a large proportion may be at risk under continued habitat degradation. The observed decline in detections of several bird species such as Ashy Prinia, Indian Robin, and Spotted Dove with increasing temperature, especially during periods of extreme heat with 8% of sites under orange heat stress<sup>79</sup> alert > 40 °C, indicates species-sensitivity to extreme heat<sup>55</sup>. These orange alert locations were in the semi-urban gradient. Both Mirzapur and Bhadohi districts experience recurring summer heatwaves<sup>79</sup>, thus enhancing green space coverage, particularly in open areas and especially at the 28 sites (13%) that are exposed to extreme heat should be prioritized in landscape management plans.

To support long-term avian biodiversity across rapidly urbanizing regions like Mirzapur and Bhadohi, urban and regional planning must prioritize the conservation and restoration of native green spaces in semi-urban gradient. Expanding the cover of native tree species such as *Butea monosperma*, *Lagerstroemia parviflora*, and *Boswellia serrata* throughout the urban matrix can enhance woodland structure, provide essential fruiting and flowering resources, and improve habitat suitability of frugivores and nectarivores. Protection of remnant natural habitats, and integrating biodiversity-sensitive design into built-up infrastructure are essential to counter habitat fragmentation and urban sprawl<sup>18</sup>. Our findings emphasize that bird species' densities are shaped by local habitat quality, along the rural–urban gradient. The association between bird guilds and specific habitat features, such as tree density for canopy foragers, water bodies for piscivores, or open fields for granivores, suggests that conserving habitat heterogeneity is key to maintaining avian diversity. Incorporating green infrastructure, regulating the spread of impermeable surface, and enhancing microhabitats through mixed-use green zones will counter extreme temperature in summer help accommodate both habitat generalists and specialists. Such integrative strategies are critical for mitigating biodiversity loss and fostering coexistence between urban development and ecological sustainability. The limitation of this study is that it represents only a temporal snapshot within a single season; to better understand population dynamics in terms of population growth rate and survival probability,

future studies should adopt open-population frameworks that incorporate multi-seasonal comparisons covering multi-year and multi-season surveys<sup>80</sup>.

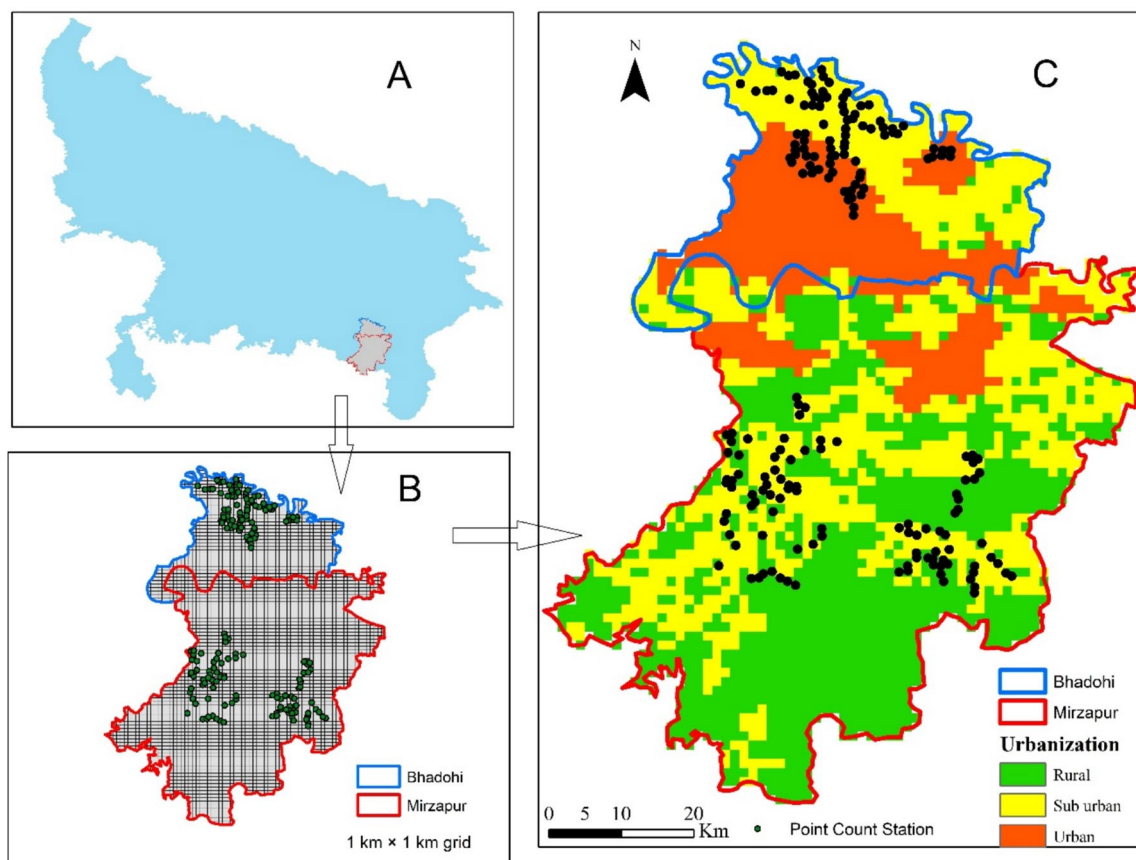
Landscape management in Mirzapur and Bhadohi districts differs substantially from that of major metropolitan cities, reflecting contrasts in governance capacity, land-use intensity, and planning priorities. In these smaller cities, landscape management is largely decentralized and incremental, with limited formal urban planning, resulting in a heterogeneous matrix of agricultural fields, fallow land, riverine corridors, village commons, and low-density built-up areas. Green spaces are often informal or multifunctional, shaped by traditional land-use practices rather than targeted biodiversity or infrastructure planning. In contrast, major metro cities operate under centralized and highly structured planning frameworks, where landscapes are intensively engineered to meet demands for housing, transportation, sanitation, and economic activity. Urban green spaces in metros are typically planned and spatially segregated, with biodiversity management often secondary to recreational or aesthetic objectives. Consequently, Mirzapur and Bhadohi retain higher landscape permeability and ecological connectivity despite limited management inputs, whereas metropolitan cities exhibit more fragmented but tightly regulated landscapes, with stronger trade-offs between development and ecological functions.

## Methodology

### Study area

Our study area is located in the Bhadohi and Mirzapur districts in South Eastern parts of Uttar Pradesh, India (Fig. 9). Both the districts are divided by the fertile plains of the Ganges River, and biogeographically lie in the Indo-Gangetic Plain. The topography of southern part is characterized by rocky hills, plateau and tropical deciduous forest patches of the Vindhyan range<sup>81</sup> with elevation up to 600 meters<sup>82</sup>. This area has rugged terrain with elevation ranging from low hills to high plateau<sup>81</sup>. The northern area is flatter and agriculturally more productive. Mirzapur (25° 9' 0" N, 82° 34' 0" E) has a total land area of 4521 square kilometres<sup>83</sup> being the largest district of the state in terms of area and Bhadohi (25.4333° N, 82.6167° E) has a total land area of 1056 km<sup>2</sup> being the smallest district in the state. The study area experiences a subtropical climate with average annual rainfall ranging from ~ 900 mm to 1200 mm (Indian Meteorological Department 2020).

We classified our study area into three landscape gradients i.e., rural, semi-urban and urban based on the Global Human Settlement Layer (GHSL)<sup>84,85</sup>. The GHSL provides a global coverage of the multi-temporal



**Fig. 8.** (A) Study area showing Uttar Pradesh administrative boundary and (B) Bhadohi and Mirzapur districts overlaid with 1 km × 1 km grids processed in ArcMap (version 10.4)<sup>104</sup>. (C) Landscape gradient showing three different urbanization gradients with 183 fixed-radius point count stations i.e. 63 in rural, 78 in semi-urban and 42 in urban as black dots.

analysis of human settlements. It combines population size, population density and grid contiguity (Table 1) and classifies the degree of urbanization at 1 km<sup>2</sup> grid cells<sup>86</sup>. We then reclassified the seven-settlement types of the original GHSL into three landscape gradients i.e., rural, semi-urban and urban (Data ID: GHS\_SMOD\_E2030\_GLOBE\_R2023A\_17). We overlaid 1 km × 1 km grid over the study area because most passerine birds in rural–urban landscapes have small home range up to 600 meters<sup>87</sup>. An adequate sampling unit as point count station was selected in each of three landscape gradients following the stratified random sampling approach<sup>63</sup>. We decided the number of point count stations in proportion to the three landscape gradients. Hence, we established a total 183 fixed-radius point count stations i.e. 63 in rural, 78 in semi-urban and 42 in urban. In cases where more than one landscape gradient occurred within a grid, we assigned the gradient based on the dominant gradient type. A minimum distance of 800 m was maintained between two adjacent point count stations during the survey.

Field methods

Avian surveys

We attempt to fulfil the assumptions of distance sampling and minimize the violations of the assumptions: objects at the point are detected with confidence, objects are detected at their initial location, and measurements are accurate. We recorded the bird species using a 100-m-fixed radius (3.14 ha) point count method<sup>87,88</sup> between March and May 2024 in summer (Supplementary Data). We conducted three repeats of each point count during the survey period amount to a total sampling effort of 1742.7 ha. The survey was performed by two-personnel team, while the observer (AG) was the same during the entire sampling period. To avoid observer error in detections, all the bird counts during the survey were conducted by the same observer (AG). We conducted avian surveys in the morning (3 h after sunrise) and evening (2 h before sunset) when the birds were active. A pair of 8 × 42 binoculars (Nikon prostaff P7) and a rangefinder (Nikon prostaff 3i) were used for bird observations and perpendicular distance measurements. We started recording avian data only 5 min after reaching the centre of the point count station. We counted all bird species seen or heard within a 100 m radius for 10 min. All individuals flying above or through our point count stations were not included in the bird counts. We recorded the detection distance bands (i.e., 0–20 m, 20–40 m, etc.) when we heard only bird calls. We randomized our surveys during point count visits for each repeat. We grouped bird species into foraging guilds following the State of India's Birds (SoIB 2023) and AVONET trait database<sup>89</sup>.

Measures of predictor variables

Earlier studies in India have shown that human activities such as passing vehicles can have negative effects on bird species<sup>90</sup>. Bird species density can vary across green spaces amidst the rural–urban matrix. We recorded twenty-three state covariates and three detection covariates at each point count station. We grouped variables into four categories; vegetation, landscape features, human activity and housing type (Table 2). All the covariate data was collected at each point count after we completed the bird counts at the time of the survey that represent state variable and three weather parameters to represent detection variable. The descriptions of all state and detection variables are explained in Table 1. The habitat parameters were measured once during the survey whereas the weather parameters were measured for each survey replicate. We recorded weather parameters (temperature, air quality index, and wind speed) during avian surveys at each point-count station for every replicate. Detection covariates were obtained using an Android phone running MIUI version 104.0.7 and the built-in weather application called AccuWeather. The application refreshes weather data at high temporal (hourly) and spatial resolution, updating conditions automatically as we moved between locations. The state variables were measured only once at each point-count station, given the brief duration of replicate surveys and the expectation that these variables would vary more spatially than temporally.

Data analysis

Assessment of survey effort and ordination of state variables

To assess the survey effort, we developed the species richness accumulation curves using the *specaccum* function in the 'vegan' package<sup>93</sup>. We selected 35 resident bird species for distance sampling analysis as these species had >25 observation frequency as per the distance sampling assumption for density modelling<sup>89</sup>. We classified bird species according to their foraging guilds; Frugivore, Nectarivore, Insectivore, Herbivore, Granivore, Omnivore, Carnivore, Scavenger and Piscivore to compare guild-level patterns of density along the rural–urban gradient.

| Urbanization nomenclature             |                               | Urbanization criteria                        |                                              |                                            |
|---------------------------------------|-------------------------------|----------------------------------------------|----------------------------------------------|--------------------------------------------|
| Nomenclature as adapted in this study | EC-GHSL Model L1 Nomenclature | Local human population density (No./sq. km.) | Cluster of human population size (Total No.) | Proportion of built-up area (% of sq. km.) |
| Rural                                 | Rural                         | < 300                                        | < 5000                                       | < 0.03                                     |
| Semi-urban                            | Urban Cluster                 | 300–1500                                     | 5000–50,000                                  | 0.03–0.5                                   |
| Urban                                 | Urban Centre                  | > 1500                                       | > 50,000                                     | > 0.5                                      |

**Table 1.** Urbanization nomenclature and criteria used in this study as adapted from European Commission-Global Human Settlements Layer (EC-GHSL) Model L1<sup>85</sup>.



| Covariates                                  | Abbreviation | Unit        | Min–Max | Mean   | SD     | Description                                                                                                                                                                                                                                 |
|---------------------------------------------|--------------|-------------|---------|--------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Detection variables</i>                  |              |             |         |        |        |                                                                                                                                                                                                                                             |
| Temperature                                 | temp         | (°C)        | 12–43   | 28.15  | 7.88   | Low temperatures indicate cool conditions and higher indicate warm conditions                                                                                                                                                               |
| Air quality index                           | aqi          | –           | 34–210  | 105.06 | 40.07  | Low AQI indicates good air quality and higher values $\geq 100$ indicates very poor air quality                                                                                                                                             |
| Wind speed                                  | ws           | Km/hr       | 0.6–68  | 12.48  | 6.75   | Wind speed during the survey recorded using the android Redmi 10 weather app                                                                                                                                                                |
| <i>Density variables</i>                    |              |             |         |        |        |                                                                                                                                                                                                                                             |
| <i>Vegetation</i>                           |              |             |         |        |        |                                                                                                                                                                                                                                             |
| Grass cover                                 | gc           | %           | 0–90    | 12.47  | 16.52  | Plants that are usually less than 30 cm in height with bud or shoot apices close to the ground <sup>91</sup> . We estimated the percentage of grass cover in 30 m radius from the centre of the point count station                         |
| Canopy cover                                | cc           | %           | 0–95    | 28.46  | 25.11  | We visually estimated the percentage of canopy cover in 30 m radius from the centre of point count station                                                                                                                                  |
| Shrub cover                                 | sc           | %           | 0–50    | 11.19  | 10.08  | We estimated the percentage of shrub cover (height: 50 cm–5 m; often with multiple stems) in 30 m radius from the centre of the point count station                                                                                         |
| Tree count                                  | tc           | Total count | 0–150   | 35.37  | 20.51  | We counted trees (height: > 5 m; DBH: > 10 cm; single well defined stem) within 100 m radius from the centre of the point count station <sup>91,92</sup>                                                                                    |
| <i>Landscape feature</i>                    |              |             |         |        |        |                                                                                                                                                                                                                                             |
| Distance to the nearest green patch         | dngp         | m           | 0–200   | 38.79  | 28.97  | We define the green patch as green spaces representing by the canopy cover with a minimum of 100 m <sup>2</sup> . We measured the closest distance from the centre of point count station to the boundary of the patch using a range finder |
| Distance to water body                      | dwb          | m           | 20–2500 | 442.31 | 375.21 | We used Google Earth to measure the distance between the centre of water body (river, stream, pond) and the point count station                                                                                                             |
| Distance to the nearest human settlement    | dhs          | m           | 5–900   | 55.33  | 98.60  | We measured the closest distance from the centre of point count station to the nearest human settlement using a range finder                                                                                                                |
| Distance to the nearest road                | dr           | m           | 5–900   | 191.01 | 196.88 | We measured the closest distance from the centre of point count station to the nearest road using a range finder                                                                                                                            |
| Area of agriculture land                    | aal          | %           | 0–97    | 37.35  | 31.26  | We used Google Earth to measure the area of agriculture land which was closest to the point count station                                                                                                                                   |
| <i>Human activities</i>                     |              |             |         |        |        |                                                                                                                                                                                                                                             |
| Area of open drain                          | aod          | %           | 0–50    | 2.98   | 6.72   | We measured the area of open drain visually within 30 m radius of the centre of the point count station                                                                                                                                     |
| Number of garbage dump sites                | gds          | –           | 0–10    | 1.04   | 1.22   | We counted the garbage dump sites within 50 m radius from the centre of the point count station                                                                                                                                             |
| Number of food shops                        | fs           | –           | 0–7     | 0.18   | 0.68   | We counted small eateries, grocery shops and food provisioning stalls within 50 m radius from the                                                                                                                                           |
| Number of houses/shops with rolling shutter | rs           | –           | 0–32    | 0.85   | 3.52   | We counted rolling shutters within 50 m radius of the point count stations                                                                                                                                                                  |
| Number of vehicles                          | vh           | –           | 0–50    | 5.77   | 8.49   | We counted the vehicles passing in 5 min duration from the point count station                                                                                                                                                              |
| Number of humans passing by                 | hm           | –           | 0–30    | 10.04  | 5.65   | We counted the number of humans passing in 5 min duration from the point count station                                                                                                                                                      |
| Number of cattle                            | cattle       | –           | 0–50    | 9.84   | 8.56   | We counted the number of cattle within 50 m radius from the point count station                                                                                                                                                             |
| Number of goats                             | goat         | –           | 0–50    | 1.89   | 5.64   | We counted the number of goats within 50 m radius of the point count station                                                                                                                                                                |
| <i>Housing type</i>                         |              |             |         |        |        |                                                                                                                                                                                                                                             |
| Number of single-story building             | ssb          | –           | 0–32    | 8.45   | 5.74   | We counted the number of single-story buildings within 50 m radius of the point count station                                                                                                                                               |
| Number of multi-story buildings             | msb          | –           | 0–20    | 1.28   | 2.51   | We counted the number of multi-story buildings within 50 m radius of the point count station                                                                                                                                                |
| Number of concrete roofs                    | cr           | –           | 0–37    | 9.68   | 6.71   | We counted the number of concrete roofs within 50 m radius of the point count station                                                                                                                                                       |
| Number of tinned roofs                      | tn           | –           | 0–15    | 4.13   | 2.59   | We counted the number of tinned roofs within 50 m radius of the point count station                                                                                                                                                         |
| Number of tiled roofs                       | tlr          | –           | 0–50    | 5.31   | 6.84   | We counted the number of tiled roofs within 50 m radius of the point count station                                                                                                                                                          |
| Number of thatched roofs                    | thr          | –           | 0–45    | 3.09   | 4.58   | We counted the number of thatched roofs within 50 m radius of the point count station                                                                                                                                                       |

**Table 2.** A summary of detection and density variables considered to describe the rural–urban gradient across point count stations. For each covariate, the table includes the corresponding abbreviation used in statistical models, the unit of measurement, the observed range (minimum–maximum), the mean value, standard deviation, and a brief description outlining the variability and scale of the covariates measured.

We z-transformed the variables using *scale* function in *dplyr* package<sup>94</sup> and then performed Pearson correlation tests among the state variables. We performed correlations using *cor* function in ‘*corplot*’ package<sup>95</sup> and *find Correlation* function in the ‘*caret*’ package<sup>96</sup>. We kept a threshold ( $\geq 0.5$ ) to retain the 19 independent state variables (Fig. S1) for further ordination and distance sampling analysis. When pairs of variables exceeded this correlation threshold, the variable with stronger ecological justification was retained<sup>97</sup>. We performed non-

metric multidimensional ordination scaling (NMDS) relating the bird abundance with the state variables using *metaMDS* function in ‘vegan’ package based on Euclidean distance method<sup>98</sup>. We fitted the NMDS ordination using *envfit* function to visualize and test the dissimilarities in the state variables along the rural–urban gradient. In addition, we created ellipse around points grouped by rural, semi-urban and urban sites. The intrinsic species which may be driving the site distribution pattern was also inspected along with habitat covariates referred to as extrinsic variables. We performed all analysis using R v4.4.1 program<sup>98</sup>.

### Detection and density modelling

We estimated group density of the birds using hierarchical distance sampling model<sup>99</sup>. This approach allows generalization of the abundance, and detection probability processes such that each can be modelled to vary with covariates by point count stations and survey occasion (replicates) and with respect to the assumptions about probability distributions<sup>100</sup>. We used multinomial-Poisson mixture model<sup>99</sup> of *distamp* function of ‘unmarked’ package<sup>101</sup> in R v4.4.1 program<sup>98</sup>. We performed model building sequentially in order to reduce the number of models to be evaluated. We defined the distance bands after exploratory analyses on the distance histograms. To calculate the detectability functions for each species, we binned perpendicular distances of birds to the observers using five distance bands (0–20, 20–40, 40–60, 60–80, 80–100 m). We first fit the intercept-only models to select the best detection function (uniform, hazard rate, half-normal, and negative exponential) for each species using the Akaike Information Criterion (AICc)<sup>102</sup>. Upon selecting the best key detection function we then tested the effect of temperature, AQI and wind speed on each species detection probability<sup>100</sup>.

We then proceeded with step-wise model selection. We first ran the univariate models with each density covariate from each sub-group (Table 2). We ran null models (i.e., without any covariates for detection or density) using the best detection function. The selection of detection model and the variable was based on AICc. The model with the lowest AICc was considered the best fit<sup>103</sup>. We then created a global model with select variables representing each group of covariates and evaluated three goodness of fit (GOF) with 1000 simulations using *parboot* function for each species (Table S1). If the goodness-of-fit (GOF) test indicated a poor fit for the global model, a multinomial-negative binomial mixture was fit using the *gdistsamp* function of ‘unmarked’ package in R v4.4.1 program<sup>99</sup> whenever there was evidence of lack of fit or overdispersion (e.g.,  $\hat{c} > 1$ ) in any global model. Models with a value of  $\Delta\text{AIC} < 2$  were averaged using *dredge* function in MuMIn package<sup>103</sup> to evaluate all the sub-models. Top models were selected using a  $\Delta\text{AIC} < 2$  threshold, and cumulative AIC weights ( $\Sigma\text{AICw}$ ) were used to evaluate the relative support for retained models. For each species we averaged the models with weights  $\geq 0.01$ . We used model-averaged coefficients from the 95% confidence model set when supporting evidence for the top-ranked model was  $< 0.95$  (i.e., the model’s weight  $w_i$ ). Finally, we used the model-averaged coefficients to predict the detection functions and density estimates. We performed Kruskal Wallis test on the difference of predicted density for each species among the landscape gradient. Then, we mapped the predicted density of each species across all the point count stations surveyed in the study area using ArcMap version 10.4<sup>104</sup>.

### Data availability

Data is provided in the supplementary material.

Received: 25 August 2025; Accepted: 13 January 2026

Published online: 01 February 2026

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## Acknowledgements

We extend our sincere thanks to the Director of Sálim Ali Centre for Ornithology and Natural History for his support throughout the study. We appreciate the efforts of Mr. Prabhat Kumar for his active participation during the field work. His support was instrumental in the successful completion of field data collection. This paper is a partial fulfilment of AG's PhD in Environmental Sciences at Bharathiar University, Coimbatore.

## Author contributions

AG: conceptualization, field survey design, field data collection, formal data analysis, visualization, first draft of the manuscript and several revisions of the manuscript. AS: writing—review and editing, feedback and critical comments on manuscript drafts. RK: methodological guidance, suggestions in data analysis and interpretation, framing research questions, manuscript review and editing; AS and RK: supervision, data analysis, critical comments on manuscript drafts, reviewing and revising the manuscript. All authors reviewed and approved the final version of the manuscript.

## Funding

This work was supported by the University Grants Commission (UGC), Govt. of India, New Delhi (NTA Ref. No.: 200510046214) to AG.

## Declarations

## Competing interests

The authors declare that they have no competing interests.

## Additional information

**Supplementary Information** The online version contains supplementary material available at <https://doi.org/10.1038/s41598-026-36510-5>.

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